Sina Abdipoor (思纳), Ph.D.

AI Engineer, Researcher, and Teacher specializing in Operations Research, Optimization, and Machine Learning

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SinaAbdipoor | Google Scholar | Sina-Abdipoor | 0000-0001-8411-5451 | Full CV

Experience

Sunway University, Lecturer, Department of Data Science and Artificial Intelligence — Dec 2025 – present
Leading an optimization research group, teaching optimization, machine learning, and software engineering, and supervising more than 20 student projects.

Universiti Putra Malaysia, Graduate Researcher & Tutor — Sep 2020 – Sep 2025
Developed hybrid metaheuristics for large university timetabling problems, and taught programming and machine learning.

Independent Consultant, Operations Research Engineer — Jan 2020 – Mar 2025
Found inefficiencies in client workflows, built and deployed optimization and prediction models, and presented the results to management.

Pars Six Sigma Excellence, Software Engineer — Jul 2017 – Jul 2019
Built and maintained a CRM system with a Java backend and a SQL database in a small software team.

Kish Institute of Science and Technology, Senior English Teacher & Supervisor — Nov 2016 – May 2019
Taught advanced English, ran IELTS workshops, and supervised and interviewed teachers.

Skills

Operations Research: Combinatorial Optimization, Mixed Integer Programming, Stochastic Optimization, Constraint Programming, Metaheuristics, Linear and Mathematical Programming, Gurobi, Google OR-Tools

Data Science: Predictive Modeling, Supervised and Unsupervised Learning, Data Visualization, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib, Power BI, Tableau, Excel

Programming: Python, Java, SQL, MySQL, SQLite, XML, Git, GitHub, UML

Languages: English (Bilingual, IELTS 8), Mandarin (Intermediate), Malay (Basic)

Projects

ESA Space Optimisation Competition (SpOC 4) — 2026
Ranked 3rd globally in the Luna Tomato Advertising challenge of the European Space Agency's space logistics optimization competition at GECCO 2026, using permutation encodings and simulated annealing. Our solutions were published as two GECCO 2026 papers.

International Timetabling Competition 2019 — 2022
An implementation of the International Timetabling Competition 2019 benchmark, the latest benchmark for university course timetabling, built as the core of my Ph.D.

Java Framework for Genetic Algorithms — 2019
A Java framework to implement evolutionary algorithms, apply them to different problems, and compare their performance. Built during my master's research.

Education

Ph.D. **Universiti Putra Malaysia** — 2020 – 2026
Computer Science
Thesis: Optimality and Generality of Metaheuristics in University Course Timetabling

M.Sc. **K. N. Toosi University of Technology** — 2016 – 2019
Computer Engineering, Artificial Intelligence and Robotics
Thesis: Analyzing Parameter Tuning of Evolutionary Algorithms in Optimization Problems

B.Sc. **Imam Khomeini International University** 2012 – 2016
Computer Engineering, Software Engineering

Selected Publications

An integration of discrete wavelet transform and time-series recurrent neural network to improve stock price prediction Aug 2026
Keong Kuon Lock, Razali Yaakob, **Sina Abdipoor**, Mingxuan Li
Applied Computing and Informatics
10.1108/ACI-08-2025-0335

Distance to Feasibility: Performance Assessment of Baseline Metaheuristics on University Course Timetabling with Student Sectioning Dec 2025
Sina Abdipoor, Mingxuan Li, Say Leng Goh, Razali Yaakob, Hao Luo, Salwani Abdullah
2025 13th IEEE Conference on Systems, Process & Control (ICSPC)
10.1109/ICSPC68261.2025.11325835

Optimality Versus Generality: Performance Assessment of Meta-Heuristics in Educational Timetabling 2025
Sina Abdipoor, Razali Yaakob, Say Leng Goh, Salwani Abdullah, Hazlina Hamdan, Khairul Azhar Kasmiran
IEEE Access
10.1109/ACCESS.2025.3565522

Hybrid Graph Neural Networks with LSTM Attention Mechanism for Recommendation Systems in MOOCs Mar 2024
Best Paper Award
Hao Luo, **Sina Abdipoor**, Qing Li, Li-Tao Xu
2024 20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)
10.1109/CSPA60979.2024.10525319

Meta-heuristic approaches for the University Course Timetabling Problem Sep 2023
Sina Abdipoor, Razali Yaakob, Say Leng Goh, Salwani Abdullah
Intelligent Systems with Applications
10.1016/j.iswa.2023.200253

Awards

Best Paper Award Aug 2026
19th International Conference on Advanced Computer Theory and Engineering (ICACTE)

National Finalist, 3 Minute Thesis (3MT) Competition Jun 2024
Universiti Kebangsaan Malaysia

Winner, 3 Minute Thesis (3MT) Competition Mar 2024
Universiti Putra Malaysia

Best Paper Award Mar 2024
20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)

Certificates

Chinese (Mandarin) Certificate May 2021
Peking University

Google Data Analytics Certificate Apr 2021
Google